Paulo Cesar Ramos Huarachi

Education

Williams College — Expected Aug 2026 – Jun 2027
M.Sc. in Policy Economics (incoming) — *Williamstown, MA, USA*
— Admitted to the Center for Development Economics. Intended focus on development economics, public policy, and applied econometrics.

Universidad Nacional de Ingeniería (UNI) — 2023 – 2025
M.Sc. in Economic Engineering (coursework completed; degree pending) — *Lima, Peru*
— Coursework completed; thesis successfully defended. Ranked 2nd in the cohort (GPA 83/100). Focus on quantitative tools for high-quality economic research. Thesis: *Effects of Fiscal Policy on Wealth Distribution: A Heterogeneous-Agent Model with Informality*.

Universidad Mayor de San Andrés (UMSA) — 2014 – 2020
B.A. in Economics, concentration in Economic Analysis — *La Paz, Bolivia*
— Quantitative track: Applied Microeconomics, Advanced Macroeconomics, Dynamic Optimization, and Model Simulation.
— Academic ranking: 4th out of 218 students (cohort). Thesis: *Effectiveness of Fiscal Policy over Bolivia's Business Cycle* (grade: 100/100).

Universidad Privada de Bolivia (UPB) — 2025 | 2022
Postgraduate studies — *La Paz, Bolivia*
— *Machine Learning and Data Science* (2025); *Quantitative Methods for Economic Analysis* (2022).

Universidad Escuela Militar de Ingeniería (EMI) — 2024 | 2017
Graduate diploma and postgraduate program — *La Paz, Bolivia*
— Graduate Diploma in Higher Education Teaching (2024). Postgraduate Program in Applied Quantitative Economics (2017).

Professional and academic experience

Ministry of Economy and Public Finance (MEFP) — 2020 – 2025
Econometric researcher, Economic & data analyst (econometrics/ML), Fiscal Analysis and Studies Unit (economic policy advisory unit) — *La Paz, Bolivia*
— Designed and executed economic and statistical analyses using macro/financial datasets to support fiscal and macro policy decisions.
— Developed forecasting and monitoring (time series, nowcasting, scenario analysis) and translated results into policy briefs, reports dashboards/figures, and presentations for senior stakeholders.
— Implemented reproducible pipelines: ETL, model estimation, and documentation for recurring reports.
— Coordinated technical work with other institutions and shared analytical results.

MEFP (Bolivia) & Inter-American Development Bank (IDB) — 2025
Econometrics & Machine-Learning Lead Analyst – Social Registry of Households Roadmap — *La Paz, Bolivia*
— Co-authored Annex 2 on household prioritization using Proxy Means Testing (PMT).
— Developed, estimated, and validated OLS and machine-learning models used in the annex.

Research Institute in Economics and Finance (RIEF) — 2022 – Present (ad hoc)
Research and Teaching Assistant (volunteer) — *Lima, Peru*
— Member of the *Heterogeneous Agents Research Initiative* and research assistant.
— Support on papers and books; transcription of handwritten notes into LaTeX; MATLAB/Python coding and preparation of figures, tables, and presentations.
— Support to academic activities and Teaching support for short courses/workshops: preparation of materials, coding labs, and problem sets, as requested.

Universidad Alberto Hurtado (Journal *Análisis Económico*) — 2024 – 2025
Ad hoc referee (peer reviewer) — *Santiago, Chile*
— Reviewed economics manuscripts and provided methodological and empirical feedback to support editorial decisions.

Universidad Mayor de San Andrés (UMSA) — 2017 – 2019
Teaching Assistant (Undergraduate) — *La Paz, Bolivia*
— Led tutorial sessions and software labs; designed and graded quantitative problem sets and exams using statistical software.

Teaching – short courses and workshops (instructor)

Universidad Nacional de Ingeniería (UNI) – FIEECS — 2025
Instructor (Master's program) — *Lima, Peru / Remote*
— Macroeconomics I and Econometrics II.

CIIP LATAM – Integral Center for Research and Professional Training — 2025 – 2026 (ongoing)
Instructor (online postgraduate programs) — *Cusco, Peru / Online*
— *Macroeconometrics and advanced macro data analysis.*
— *Development econometrics and panel data analysis: models for decision-making.*
— *Bayesian econometrics* and *econometrics for forecasting with machine learning and deep learning.*

RIEF – Advanced Economics Program for Provincial Universities — 2025
Instructor – Econometrics II: nonlinear time series — *Lima, Peru / Virtual*
— Nonlinear time series; regime-switching models and ML-based regime classification in macroeconomic applications.

RIEF Summer School in Economics and Finance (5TH–6TH) 2023 – 2024
Boot Camp – intensive MATLAB *Lima, Peru*
— Programming fundamentals in MATLAB; data handling and visualization with applications in economics and finance.

Universidad Mayor de San Andrés (UMSA) – Department of Economics 2017 – 2019
Teaching assistant (undergraduate) *La Paz, Bolivia*
— Microeconomics I, Macroeconomics I, and Macroeconomics II.

Advanced training – courses (participant)

Certificate Program in Applied Generative AI 2025
Johns Hopkins University – Whiting School of Engineering *Online, U.S.*
— Certificate program covering Python, generative AI, *prompt engineering*, transformers, RAG, and *fine-tuning*.
— Faculty includes Ian McCulloh, Iain Cruickshank, and Pedro Rodríguez (Johns Hopkins Applied Physics Lab).

São Paulo School of Advanced Science on High-Dimensional Modeling (FGV EESP) 2025
Selected participant (FAPESP funding) *São Paulo, Brazil*
— Topics: high-dimensional forecasting; ML for portfolio allocation; climate econometrics; crypto-finance; dynamic factor models.

MEFP – Inter-American Development Bank (FISLAC) technical assistance 2024
Macro-fiscal assistance and general equilibrium modeling *La Paz, Bolivia*
— Fiscal sustainability, macro-fiscal impact analysis, and general equilibrium models for policy design and forecasting.

Grupo Lambda Perú 2023 – 2024
Training program *Lima, Peru*
— Macro analysis and forecasting with ML; applied macroeconometrics; R and Python for economists.

IMF (edX) & Coursera 2019 – 2026
Certificates *Online*
— IMF (edX): *Monetary Policy Analysis & Forecasting*; *Macroeconomic Diagnostics*; *Financial Programming*; *Macroeconometric Forecasting*.
— Coursera: *Supervised Machine Learning: Regression and Classification*; *Advanced Learning Algorithms*; *Natural Language Processing with Classification and Vector Spaces*; *Harnessing the Power of Data with Power BI*; *Visualization for Data Analysis with Power BI*; *Game Theory with Python*; *Business English (Finance & Economics)*.

Conferences and academic workshops

IDEAs–IIR Advance Student Workshop on Development Challenges (IIR–UWI) Oct. 2025
Selected participant *St. Augustine, Trinidad and Tobago*
— Workshop: *External Vulnerability and the Development Challenge in the Current Conjuncture*, The University of the West Indies.
— Participation in the IDEAs–IIR international conference on *Caribbean Unorthodox Economic Thought*.

Awards and honors

2025 MEFP & IDB: recognition for contributions in econometrics and machine learning to the Social Registry of Households Roadmap.

2024 Central Bank of Bolivia: 2nd place for an article on the macroeconomic and redistributive impacts of taxes (with A. Monzón).

2023, 2025 MEFP: academic recognition certificate (signed by the Minister) for research contributions.

2021–2024 MEFP: recognition for outstanding performance and teamwork.

2019 Excellence Course in Economics (Univ. of Chile, UPB, UCB): completed with distinction; 3rd place.

2017 UMSA Department of Economics: invited to serve as teaching assistant in Microeconomics.

Publications

— Heterogeneous Effects of Fiscal Policy on Wealth Distribution: A Heterogeneous Agents Approach in Emerging Economies with High Informality — 2025 — *Master's thesis*; with Ian Carrasco; Advisor: Hamilton Galindo, Ph.D.
— Fiscal vs. Monetary Dominance in Bolivia (1930–2024): A Historical Index from Central Bank Reports Using NLP and Numerical Extraction — 2026 — *Working paper (in progress).*
— Bank Exposure to Financial Risk and Monetary Policy Transmission: Evidence from Bolivia and Peru — 2025 — *Working paper (in progress).*
— Assessing the Effectiveness of Monetary Policy Tools Under De-dollarization: A Time-Varying Analysis — 2024 — *Working paper (in progress).*
— Machine Learning Insights into Bolivia's Economic Downturns — 2024 — *Article (accepted/published).*
— Macroeconomic and Redistributive Impact of Taxes — 2024 — *Published report, Central Bank of Bolivia*; with A. Monzón.
— Assessment of Household Welfare in Food Consumption: An Application of Vartia's Algorithm — 2023 — *Journal article, CIEB Journal.*
— Informality in Labor Markets — 2020 — *Book chapter, Konrad Adenauer Stiftung (KAS).*
— Small Open Economy Model: The Case of Bolivia — 2019 — *Working paper, REDES (UCB) – No. 5.*

Programming and tools MATLAB, Python, Stata, EViews, R; VS Code, Jupyter, Git/GitHub, Microsoft Office.

Econometrics and modeling Classical and Bayesian econometrics, DSGE/Dynare, IRIS and BEAR toolbox; *machine learning* methods and algorithms.

Research workflow LaTeX (papers, Beamer, TikZ, BibTeX); Overleaf/TeXStudio/WinEdt; literature managers and tools: Zotero, Connected Papers, Research Rabbit.

Languages Spanish (native), English (advanced reading and academic writing).

Other activities

2022 UMSA: member of the Graduates Commission for ARCUSUR (MERCOSUR) accreditation.

2022 RIEF: participant in the *EconFin Reading Group* (Heterogeneous Agents in Asset Pricing).

2018–2019 UMSA: member of the Merit Evaluation Committee (faculty; research and service assistants).

2018 UMSA: representative of teaching assistants.